

Two-Track DCA_{xy} Comparison

FastSim closest-point residual vs. cross-product projection

Definitions

For track PCA positions $\mathbf{r}_i, \mathbf{r}_j$ and momentum directions $\mathbf{u} = \mathbf{p}_i/|\mathbf{p}_i|, \mathbf{v} = \mathbf{p}_j/|\mathbf{p}_j|$, define

$$\mathbf{n} = \frac{\mathbf{u} \times \mathbf{v}}{|\mathbf{u} \times \mathbf{v}|}, \quad \Delta \mathbf{r} = \mathbf{r}_j - \mathbf{r}_i.$$

The cross-product method discussed here is

$$D_{xy}^{\text{cross}} = \mathbf{n} \cdot (\Delta x, \Delta y, 0).$$

This is a projection of the transverse PCA separation onto the normal vector of the two track directions.

The current FastSim method first solves the two closest points $\mathbf{c}_i$ and $\mathbf{c}_j$ on the full 3D track lines, then computes

$$\boldsymbol{\rho} = \mathbf{c}_i - \mathbf{c}_j, \quad D_{xy}^{\text{FS}} = s \sqrt{\rho_x^2 + \rho_y^2},$$

where the sign s is set from

$$s = \text{sign} [((\mathbf{p}_{T,i} + \mathbf{p}_{T,j}) \times (\mathbf{r}_i - \mathbf{r}_j))_z].$$

Main Result

These are not the same observable. The FastSim value is the transverse magnitude of the closest-point residual. The cross-product value is a normal-vector projection after forcing the PCA separation's z component to zero. They can have similar overall scale, but they are not related by a single event-by-event scale factor.

Signed values should not be compared directly without first harmonizing the sign convention. The table below compares absolute magnitudes.

Numerical Check

The following check used the default FastSim configuration, seed 20260509, 20,000 generated events, and 12,919 selected Si-Si pairs. All distances are in cm.

Quantity	Median	68%	90%	95%	99%	Max
$ D_{xy}^{\text{FS}} $	0.00500	0.00819	0.01683	0.02260	0.03636	0.08340
$ D_{xy}^{\text{cross}} $	0.00419	0.00710	0.01578	0.02174	0.03611	0.08151
$ D_{xy}^{\text{cross}} - D_{xy}^{\text{FS}} $	0.00240	0.00400	0.00844	0.01159	0.01925	0.03683

The ratio $|D_{xy}^{\text{cross}}|/|D_{xy}^{\text{FS}}|$ had median 0.91. Its 5–95% interval was 0.087–6.64, with a 99% value of 32.7. The large upper tail appears when the FastSim denominator is close to zero.

Interpretation

For resolution-width comparisons, the two definitions may look broadly similar because both are controlled by the same PCA and angular scales. For event-by-event comparisons, DCA cuts, or tail studies, the difference can be substantial. A few 10^{-3} cm is typical in the default toy sample, while the tail reaches a few 10^{-2} cm.